

XAI Experience Quality (XEQ) Scale Handbook

V.1.0.0

Introduction

The XAI Experience Quality (XEQ) Scale was developed to help researchers and practitioners evaluate how users experience AI explanations — especially in interactive, multi-step settings. While the original research provides the theoretical foundation and validation, this handbook serves as a practical guide to applying the scale.

Our aim is to make the XEQ Scale easy to use in real-world projects. This handbook summarises the essentials: what the scale measures, how to apply it, and how to interpret the results. Where relevant, we point to further resources for those who wish to explore the research in more depth.

Construction of the XEQ

The XEQ Scale was developed to assess the quality of user experiences with explainable AI systems, particularly those involving interactive, multi-shot explanations. The construction process followed established practices from psychometric theory to ensure both rigour and practical value.

An initial pool of 40 questionnaire items was compiled through a targeted literature review and expert brainstorming. These items were sourced from existing evaluation tools in XAI and HCI and extended with new items to reflect the interactive and user-centred nature of XAI experiences.

The item pool was refined through a two-stage expert review process and then evaluated in a content validity study involving 13 XAI researchers and practitioners. Based on expert feedback, the scale was reduced to 18 items, grouped into four key dimensions: **Learning**, **Utility**, **Fulfilment**, and **Engagement**.

A large-scale pilot study with 238 participants was conducted to validate the scale. Statistical analyses confirmed strong internal consistency (Cronbach's $\alpha = 0.96$), discriminant validity, and test-retest reliability. Factor analysis further supported the four-dimensional structure of the scale and identified an overarching single factor representing overall XAI experience quality.

For detailed information on the construction and validation of the XEQ Scale, see: Wijekoon, Anjana, et al. "XEQ Scale for Evaluating XAI Experience Quality." *arXiv preprint arXiv:2407.10662* (2024).

Scale Structure

The XEQ Scale contains **4 dimensions with 18 items**. Each dimension reflects a distinct aspect of user experience when interacting with explainable AI (XAI) systems:

- **Learning** (4 items): Assesses how well the XAI experience improves the user's understanding of the system and its decisions.
- **Utility** (6 items): Measures how effectively the experience supports the user in achieving their goals or completing tasks using the AI system.
- **Fulfilment** (5 items): Captures the extent to which the XAI experience meets the user's expectations and explanation needs.
- **Engagement** (3 items): Evaluates the interactivity, clarity, and overall smoothness of the explanation process.

All items are rated on a 5-point Likert scale, from *Strongly Disagree* to *Strongly Agree*. These dimensions allow for both an overall XAI Experience Quality (XEQ) score and more detailed insights through per-dimension factor scores.

Can I Remove Items?

You **cannot remove individual items** from the questionnaire, as each one contributes to the reliability of its corresponding XEQ factor. However, if a particular **factor** (e.g., Engagement or Learning) is not relevant to your context, you may choose to **exclude the entire factor** from your evaluation. This allows you to tailor the tool to your needs while still preserving the validity of the remaining results.

Typical Application Scenarios

Test if a system has sufficient quality

This scenario uses XEQ to assess the overall quality of a system. The figure below shows **factor scores** across the four XEQ dimensions—**Learning, Utility, Fulfilment, and Engagement**—and they are all **roughly equal and above 3**. This indicates that the system provides a **balanced experience**, with no major weaknesses across dimensions. Such a pattern suggests that the product is likely to meet user expectations broadly and consistently, making it a suitable candidate for wider deployment or continued use. However, there is still room for improvements.

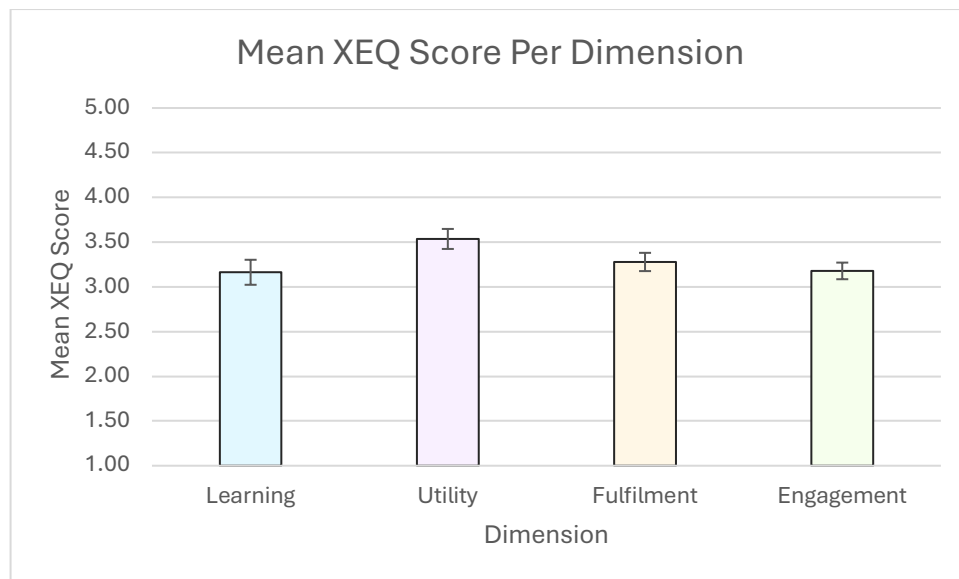


Figure 1: Example Factor Scores

Better personalise explanation experiences

This scenario uses XEQ to identify differences in experience quality across stakeholder groups. The figure displays **stakeholder scores**, revealing that **Admin Assistants had the lowest ratings**, with **all scores below 3** and **Learning as low as 1.5**. This suggests that the current system or explanation approach does not meet their needs—particularly in helping them understand how or why the system works. These insights highlight the importance of **personalising explanations** based on role-specific expectations and background knowledge, ensuring that all users—not just some—can benefit meaningfully from the system.

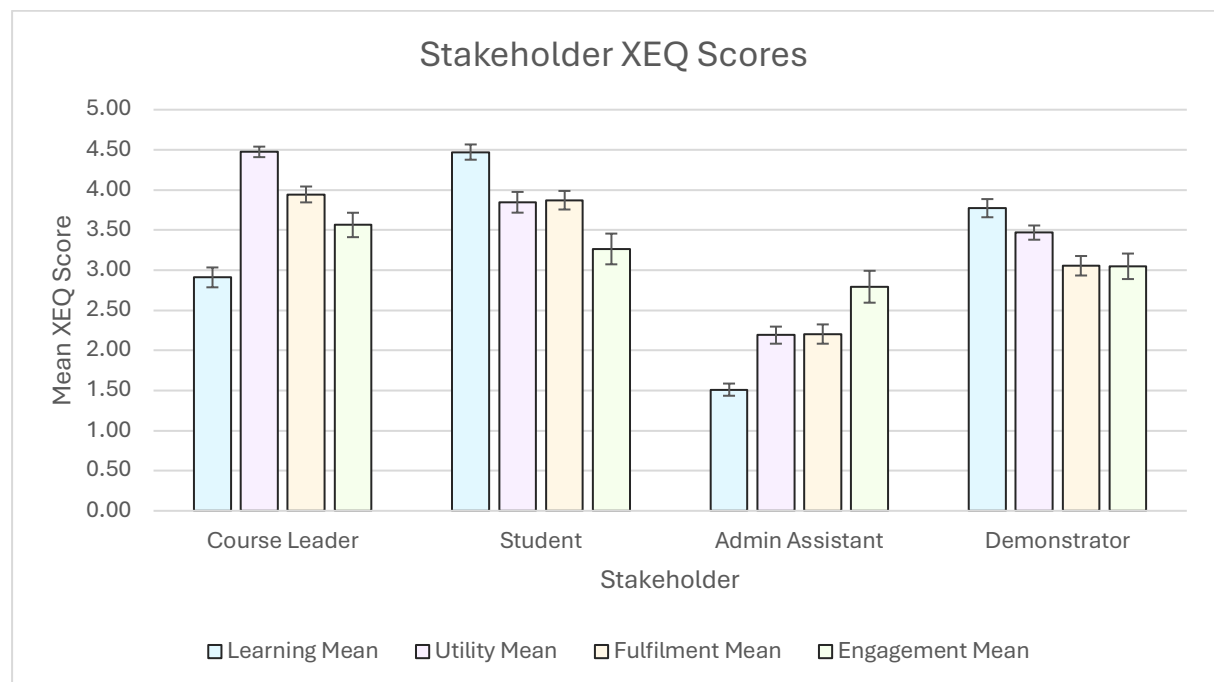


Figure 2: Example Stakeholder Scores

Compare competing XAI systems

XEQ can be used to compare the user experience quality of **two or more AI systems** performing similar tasks—for example, evaluating whether a new version of a model offers clearer explanations or is more engaging than an existing one. By collecting XEQ responses for each system separately, you can compare scores across dimensions and stakeholder groups to identify which system delivers a better overall experience or meets specific user needs more effectively. While the current tool supports analysis of **one system at a time**, we plan to release a **comparison sheet** in the near future to make side-by-side evaluations easier and more structured.

Benchmarking

XEQ is designed to support **benchmarking** by enabling organisations to track experience quality over time or compare results across systems and domains. Although this functionality is conceptually supported, **no benchmark data is available yet**, as the tool has only just been released. As more users adopt XEQ and share results, we plan to develop a **benchmark dataset** to allow comparisons against typical experience scores. For now, you can begin building your own internal benchmarks by applying XEQ consistently across projects and tracking changes over time.

Translation to Other Languages

At present, the XEQ Scale is only available in English. However, we are actively working on translating it into other languages to make it more accessible to a global audience. Translations will be added as they become available, and we welcome community contributions. If you are interested in helping with a translation, please get in touch.

XEQ Hardcopy

A **printable hardcopy version** of the XEQ questionnaire is available for download from the [XEQ website](#). This format is ideal for use in interviews, workshops, or situations where digital access is limited. Responses collected on paper can later be entered into the Excel tool for analysis.

Online Application

The online version of the XEQ tool is designed for **individual use**—it allows a single user to complete the questionnaire and reflect on their experience. Once completed, users can **print out their responses to each item**, which can then be manually entered into the XEQ Excel tool for group-level analysis and visualisation. This is especially helpful for workshops, pilot testing, or gathering initial feedback. In future, we plan to develop an **online results collation tool** to make it easier to collect and analyse responses from multiple users automatically.

How to Use the Excel Tool?

The XEQ Excel Tool is designed to simplify the process of scoring and analysing responses to the XAI Experience Quality (XEQ) Scale. To use it, simply enter participant responses into the 'Data' sheet and define the stakeholders who you are interacting with your AI system in the 'Stakeholders' sheet within the file XEQ_Data_Analysis_Tool.xlsx.

All key metrics — including stakeholder scores, dimension-level factor scores, and the overall system XEQ score — are automatically calculated. You do not need to perform any manual analysis or write formulas.

The example illustrates how XEQ scores can be used to evaluate perceptions of a single system across different stakeholder groups. While the current tool supports rich analysis of one system at a time, **it does not yet support direct comparisons between multiple systems or versions** (e.g., comparing version A vs. version B).

A separate comparison tool—**XEQ_Compare_Systems.xlsx**—is **planned for future release** to support such analyses. In the meantime, users can still manually compare outputs by placing results side by side and noting differences in scores and stakeholder patterns.

Please make sure you are using the **latest version** of the XEQ Excel Tool, available at www.xeqscale.org, as it is actively maintained and improved based on ongoing research and user feedback.

How to Interpret the Data

How do I interpret the scores?

The XEQ tool provides three layers of results to help you understand user experience from different perspectives:

- **Stakeholder Scores**
These show how each group of users rated their experience. For example, if *students* give high scores across all four dimensions but *admin staff* score much lower, it suggests the product works well for learners but may not meet the needs of administrative roles. This helps identify whose experience is strong and where improvements may be needed.
- **System Score**
This is the overall average across all stakeholder groups and all XEQ dimensions. For instance, a system score of **4.2** indicates that, on average, users had a very positive experience. If this number is closer to **3.0**, the product may still need refinement. Think of it as a high-level snapshot of quality across the board.

- **Factor Scores (Learning, Utility, Fulfilment, Engagement)**

These tell you how the system performs in each of the four XEQ dimensions. For example, if **Utility** scores are high (e.g., 4.5), it means users found the product effective and functional. But if **Engagement** scores are low (e.g., 2.8), the system may be perceived as boring or difficult to interact with. This highlights specific areas where experience can be improved.

Taken together, these scores help you answer:

- *Who is having a good (or poor) experience?* (Stakeholder scores)
- *How good is the overall experience?* (System score)
- *Which aspects of the experience stand out or fall short?* (Factor scores)

As more projects use the XEQ tool, we plan to develop a **benchmark dataset** to support comparisons. This will allow you to see how your scores compare to typical ranges in similar systems, making interpretation even more meaningful and grounded in real-world performance. **For now, a simple rule of thumb is to interpret scores below 3 as needing improvement, and scores above 3 as acceptable or positive.**

What do the error bars mean?

In the XEQ tool, each mean score is shown with an accompanying **error bar**, which visually represents the **confidence interval** for that score. By default, this is set to **95%**, but users can adjust the confidence level based on their needs.

Think of it this way: if you were to repeat the same evaluation many times under the same conditions, small variations would naturally occur due to randomness. The error bar marks the range in which **95% of those repeated mean scores** would be expected to fall. In other words, it shows how precisely your result reflects the underlying truth.

Two main factors influence the size of the error bar:

- **Sample size:** Larger participant numbers tend to reduce the width of the interval.
- **Response consistency:** When participants agree more closely in their answers, the interval also becomes narrower.

Wider error bars suggest greater uncertainty in your measurements, meaning you should be more cautious in interpreting the results. Often, a large interval is a sign of too few responses. In such cases, gathering more data is advisable to improve the reliability of your findings.

How to interpret the standard deviation

Standard deviation is a measure of how much individual responses vary around the average score. A **low standard deviation** means that most people gave similar ratings,

indicating agreement or consistency in their experience. A **high standard deviation**, on the other hand, suggests that responses were more spread out—some rated the experience highly, while others rated it much lower. This can be a sign of differing expectations, inconsistent performance, or variation in how different users interacted with the product.

Frequently Asked Questions (FAQs)

For answers to common questions and tips on getting started, please check the FAQ section on our website. As the XEQ tool has just been released, the FAQ is still being developed—but we'll be adding new entries based on user feedback and common queries. Stay tuned!